

Final Year Capstone Projects

Two Project Proposals

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Two Exciting Opportunities

Topic 1: Applied

Campus Navigation Web App

Topic 2: Theory

Information Theoretic Inequalities

Topic 1

Campus Light Bus Navigation App

The Problem

- CUHK campus runs paid light bus service for visitors
- First-time visitors lack clear stop instructions
- Passengers miss their stops frequently
- Confusion leads to poor user experience

Proposed Solution

- Real-time location tracking web application
- User selects destination from dropdown menu
- App alerts user at correct stop
- Walking directions to final destination

Key Technical Components

- GPS-based real-time tracking system
- Interactive map interface with route display
- Proximity alerts and notifications
- Integration with campus map data

Student Requirements

- Design and develop complete web application
- Test with actual campus visitors
- Software development skills (or willingness to learn)
- Experience with web technologies preferred

Topic 2

Information Theoretic Inequalities

Prove information theoretic inequalities through rigorous mathematical analysis

- Work closely with me or doctoral students
- Develop deep theoretical understanding
- Contribute to foundational research

Network Information Theory

- Interference channels and capacity regions
- Multi-user communication systems
- Entropy and mutual information bounds

Additive Combinatorics

- Sumset inequalities and entropy
- Formal equivalence theorems
- Bridge between combinatorics and information theory

Functional Analysis

- Gaussian optimization techniques
- Extremality and variational methods
- De Bruijn and Fisher information inequalities

Student Requirements

- Strong mathematical background
- Interest in theoretical research
- Comfort with rigorous proofs
- Willingness to engage deeply with literature

Project Comparison

Applied Project

Software development, real-world testing, user experience focus

Theory Project

Mathematical proofs, research mentorship, foundational contributions

Questions?

Choose the project that matches your interests!

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